

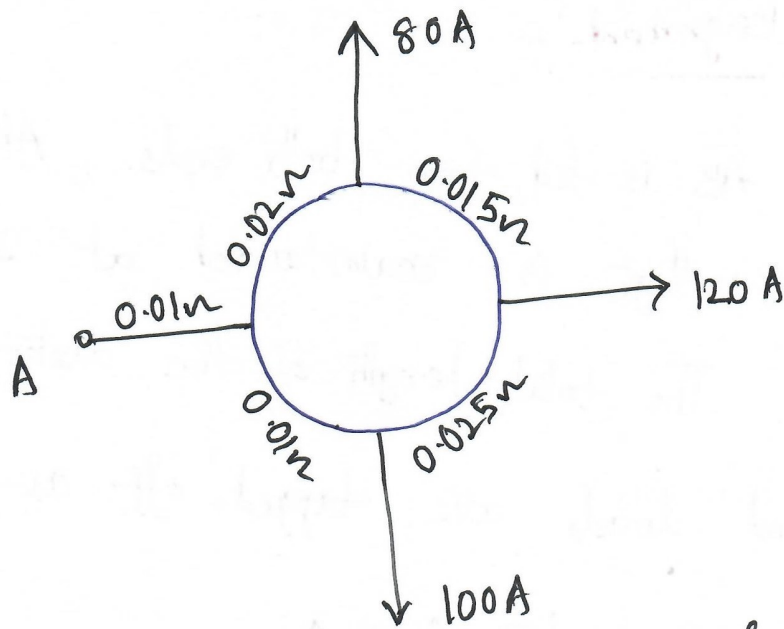
Assignment-4

Q. A distributor AB is fed from both ends. At feeding point A, the voltage is maintained at 236 V and at B 237 V. The total length of the distributor is 200 meters and loads are tapped off as under

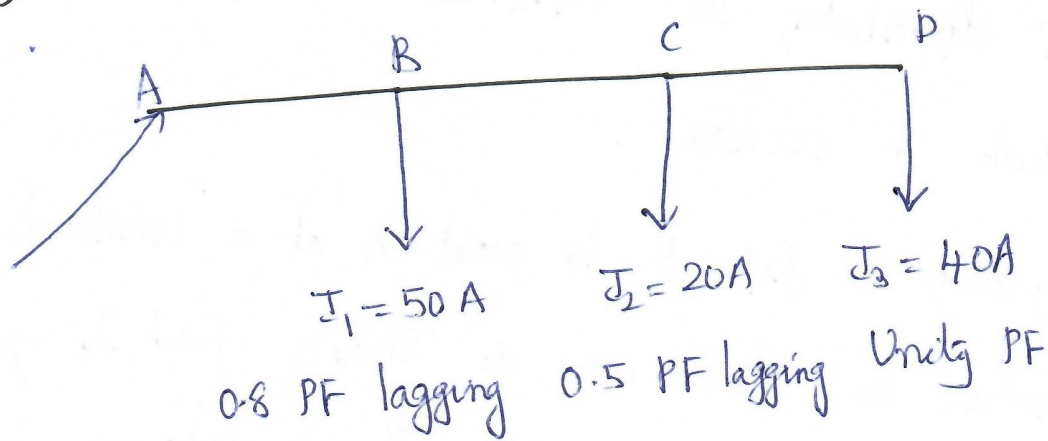
- (i) 20 A at 50 meters from A
- (ii) 40 A at 75 meters from A
- (iii) 25 A at 100 meters from A
- (iv) 30 A at 150 meters from A

The resistance per km of one conductor is 0.04Ω . Calculate the currents in the various sections of the distributor, the minimum voltage and the point at which it occurs.

A ring main is fed at the point A at a constant voltage of 240 volts. The loads are as shown. Find the potential difference at each load point. The resistances given are those of each conductor.



The loading on a distributor is shown in below figure. The distributor is a two core cable for which the resistance and reactance are 0.25Ω and 0.125Ω per 1,000 meters of cable run respectively. What should be the voltage at the point A to maintain 400V at point D?



4. A 3- ϕ ring main ABCD, fed from the end A supplies balanced loads of 50A at 0.8 pf lagging at B, 120A at unity pf at C and 70A at 0.866 pf lagging at D, the load currents being referred to the voltage at point A.

The impedances per phase of the various line sections are:

$$\text{Section AB} = (1 + j0.6) \Omega$$

$$\text{Section BC} = (1.2 + j0.9) \Omega$$

$$\text{Section CD} = (0.8 + j0.5) \Omega$$

$$\text{Section DA} = (3 + j2) \Omega$$

Determine the currents in the various

sections.

5. Calculate voltage at a point 300 metres from the feeding point maintained at 250 volts when the cable is 1,000 metres long, has a resistance of 0.05Ω per km for each conductor and supplies a uniform load of 0.5 A/metre .

find also the voltage at this point when the cable is supplied at 250 V at both ends.